



IILM UNIVERSITY, GREATER NOIDA, U.P.
School of Computer Science and Engineering

INTERNSHIP REPORT
ON
“WILDLIFE CONFLICT AND MOVEMENT PREDICTOR
USING AI”

Submitted to
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.
In partial fulfilment of the requirement for the degree of
B.Tech CSE

By
Roll Number: 25SCS1003001223
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Batch: 2026-27
Section: 2CSE31

Internship Organization: MARS Internship
Internship Period: 1 June 2026 – 15 July 2026

CANDIDATE'S DECLARATION

I, Abhishek Kumar Mishra, hereby declare that the internship report titled "Wildlife Conflict and Movement Predictor using AI" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The report presents my internship experience, my contribution to the team project, and the development work carried out during the internship period. References used in preparing this report have been acknowledged appropriately. I affirm that this report has not been submitted to any other institute for the fulfilment of any other degree or diploma requirement.

Signature: _____

Student Name: Abhishek Kumar Mishra

Roll No.: 25SCS1003001223

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported me during my summer internship and the development of the project "Wildlife Conflict and Movement Predictor using AI." The internship provided me with an opportunity to participate in a practical, team-based software project and to understand how a project can be developed around a real-world social and environmental problem. I am thankful to IILM University, School of Computer Science and Engineering, for providing the academic framework and opportunity to undertake the internship as part of the B.Tech CSE curriculum. I am also grateful to the MARS Internship program for providing the environment in which our team could work on the project remotely and collaboratively.

I would like to acknowledge my teammates for their cooperation throughout the internship. Since the project was developed as a team effort, the final system reflects the combined work of all members. My own contribution was mainly focused on backend development, major frontend improvements, and participation in the overall system development.

Finally, I express my gratitude to my parents and everyone who encouraged and supported me throughout the internship period.

Signature: _____

Student Name: Abhishek Kumar Mishra

Roll No.: 25SCS1003001223

INTERNSHIP COMPLETION CERTIFICATE



CERTIFICATE OF INTERNSHIP COMPLETION

Reference No.: MARS/INT-CERT/2026/055

Date of Issue: July 31, 2026

This is to certify that

Abhishek Kumar Mishra

has successfully completed an internship with MARS – Centre for Sustainable Community Development from June 1, 2026 to July 15, 2026, contributing sincerely to the organisation's initiatives in sustainable community development. During this period, the intern demonstrated dedication, a strong learning attitude, and a genuine commitment to the values of sustainability and community service.

Sustainability is not a destination but a continuing journey. May the knowledge and experience gained during this internship inspire you to keep championing sustainable practices, and to carry this commitment forward into every community and career you touch in the years ahead.

Dr. Manish Prateek

Director

MARS – Centre for Sustainable
Community Development



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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

Wildlife movement near human settlements and the resulting human–wildlife conflict can create risks for both local communities and animals. People living close to forest boundaries may encounter wild animals while carrying out routine activities, while rangers and park authorities need timely information to understand where animal movement and conflict are occurring. The project undertaken during the MARS Internship was designed as a web-based platform to support these needs through a combination of live mapping, incident reporting, analytics, dataset management and AI-assisted risk prediction. The project was titled “Wildlife Conflict and Movement Predictor using AI.” The platform was designed for two broad categories of users: people living in rural areas around wildlife habitats and rangers or personnel associated with national parks. The central idea was to provide a single interface through which relevant wildlife movement information and conflict-related information could be viewed and used for preventive decision-making.

Rather than limiting the system to a single protected area, the developed platform was structured to support information from ten national parks. A park-switching capability allows the user to change the selected park and work with the corresponding information. This makes the platform more flexible and demonstrates how the same system structure can be extended to multiple wildlife regions.

A major feature of the platform is its use of structured datasets. New information received from national parks can be uploaded through the Dataset Management module instead of requiring the website to be manually rebuilt every time new data becomes available. The data-management interface provides a controlled way to introduce new data into the platform and maintain information that can later support analytics and prediction.

The project was developed as a collaborative team effort during an online internship. My contribution was substantial across the overall development of the system, with primary involvement in backend implementation and major frontend modifications. The report therefore focuses on the complete system while clearly presenting my role as one part of the collaborative development process.

Item	Details
Project Title	Wildlife Conflict and Movement Predictor using AI
Internship Organization	MARS Internship

Internship Mode	Online
Internship Period	1 June 2026 – 15 July 2026
Project Coverage	10 national parks
Primary Users	Rural communities and national-park rangers
Project Type	Web-based wildlife monitoring and risk- support platform
My Contribution	Overall development contribution; major backend and frontend involvement

4.2 Organization Profile

MARS Internship was the organization/program through which the internship was conducted. The internship was completed online and involved collaborative project development by a team of students. The central practical activity of the internship was the development of a working software project around the selected wildlife- conflict problem. During the internship, the team worked remotely and coordinated the development of the different parts of the application. The work environment emphasized project completion, collaboration, implementation and iterative improvement of the developed system. The internship therefore provided exposure to the process of taking a project idea and turning it into an integrated application with multiple user-facing and data- oriented modules.

The project work was not limited to creating static screens. The team developed a connected system in which the frontend interfaces, backend functionality, stored information, datasets and prediction-oriented components were intended to work together. This made the internship a practical software-development experience rather than only a theoretical exercise.

Because the information available for this report is primarily based on the internship project and its demonstration, this section intentionally avoids adding unsupported claims about the internal corporate structure, history or business operations of MARS Internship.

4.3 Problem Statement

Human–wildlife conflict becomes difficult to manage when information about animal movement, sightings, conflict incidents and risk areas is scattered, delayed or difficult to interpret. Rural communities may not

know which areas currently have a higher likelihood of wildlife movement, while park rangers may need to examine different sources of information before deciding where attention should be concentrated. A second challenge is that wildlife information changes over time. New observations and incident records can become available from national parks, meaning that a useful monitoring platform must provide a practical mechanism for updating its underlying datasets. A system that cannot easily accept new information may become outdated even if its initial dataset is accurate.

The project therefore addresses the need for an integrated platform that can combine wildlife movement information, incident reporting, analytics, data management and AI-assisted risk information in one interface. The intended result is not to replace the judgement of rangers or authorities, but to provide information that can support awareness, monitoring and preventive action.

4.4 Project Objectives

The main objectives of the project were to create a centralized and usable platform for monitoring wildlife-related information and supporting responses to potential conflict situations.

No.	Objective
1	Provide a web-based interface for accessing wildlife movement and conflict-related information.
2	Allow users to view wildlife activity and relevant locations through a map-based interface.
3	Provide analytics that help users understand sightings, threat levels and activity patterns.
4	Provide a structured mechanism for reporting wildlife sightings or conflict incidents.
5	Provide AI-assisted risk information to indicate periods or areas that may require greater attention.
6	Allow new datasets from national parks to be uploaded and managed through the system.

7	Support multiple national parks through a park- selection mechanism.
8	Create a practical interface that can support both community awareness and ranger decision- making.

4.5 Scope of the Project

The scope of the project covers the development of a web platform that brings together several information and decision-support functions related to wildlife movement and human–wildlife conflict. The system was demonstrated with information covering ten national parks and was designed so that the selected park could be changed within the application.

Area	Scope Included
Wildlife Monitoring	Viewing animal movement/location information through the live map interface.
Conflict Awareness	Displaying conflict-related zones, incidents and risk information.
Incident Reporting	Recording wildlife sightings or conflict incidents with relevant details.
Analytics	Viewing statistical and visual information about wildlife activity and threat patterns.
AI Support	Presenting risk-prediction information and recommended attention/action areas.
Data Management	Uploading and managing new datasets received from national parks.
Multi- Park Support	Switching between supported national parks within the platform.
User Support	Providing information that can assist rural communities and park rangers.

Out of Scope	The system is not presented as a replacement for official wildlife authorities, field verification or emergency response systems.
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4.6 Technologies and Tools Used

The project was implemented as a web application using JavaScript-based development and a MongoDB data store where required. The project also included an AI-oriented component for generating wildlife risk information. Development involved iterative implementation and modification of both the backend and frontend portions of the system.

Technology / Tool	Use in the Project
JavaScript	Primary programming language used in developing the web application.
MongoDB	Used for storing and managing application data where a database was required.
Web Frontend	Used to provide dashboards, maps, forms, analytics and other user-facing interfaces.
Backend Services	Used to handle application-side processing, data operations and communication between the interface and stored information.
AI Risk Component	Used to provide risk-oriented wildlife information and prediction-related outputs.
Dataset Management	Used to upload and maintain incoming park datasets so that the platform can be updated.
Development / AI-assisted tooling	Used during development to accelerate implementation and iteration of the application.

The report intentionally describes the technologies at the level supported by the project material. Specific libraries or frameworks are not listed as confirmed technologies unless they were explicitly identified during the internship.

4.7 System Architecture

The system follows a modular web-application structure in which users interact with the frontend interface, the frontend communicates with backend services, and the backend works with stored datasets and application data. The AI-oriented risk component operates as a decision-support layer using the information made available to the system. The architecture is intended to keep the different responsibilities of the platform organized while allowing new park data to be introduced over time.

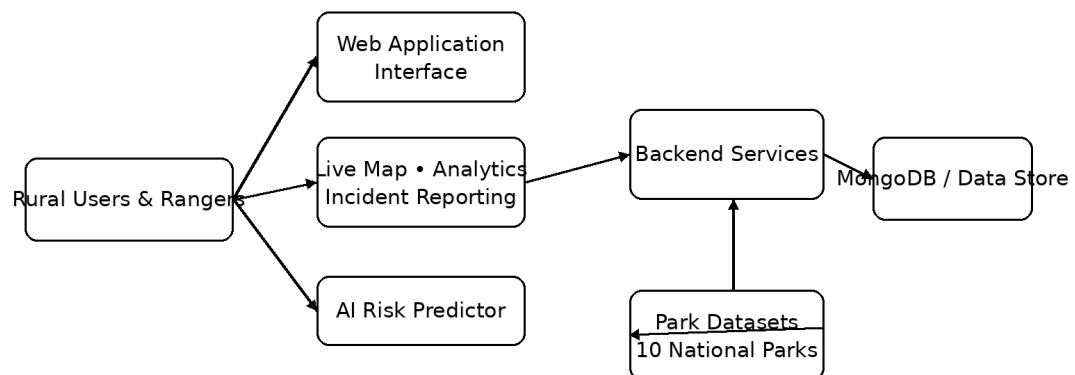


Figure 4.1: Overall System Architecture

At the user level, rural users and rangers access the web application through the available interface. The application provides the live map, analytics, incident reporting, risk prediction and dataset-management functions. The backend acts as the central processing layer for application operations and data access. MongoDB provides persistent storage for information that needs to be retained. Park datasets form an important input to the system because new information can be uploaded and managed without rebuilding the complete interface. The architecture also reflects the project's multi-park design. Instead of creating a separate application for every park, the same application can be used with different park datasets. The park-switching mechanism allows the user to select the required park and view the corresponding information.

Major System Modules

Module	Purpose

Live Map Dashboard	Provides a map-based view of wildlife locations, movement-related information, park boundaries and conflict/risk areas.
Incident Reporting	Provides a form-based mechanism for recording wildlife sightings and conflict incidents.
AI Risk Predictor	Provides risk-oriented information such as risk periods, conflict categories, hotspots and movement/activity guidance.
Dataset Management	Allows new datasets to be uploaded, checked and managed for continued use by the platform.
Park Selection	Allows the user to switch between supported national parks.

4.8 Methodology

The project methodology followed an iterative development approach. The team first identified the wildlife-conflict problem and the type of information that would be useful to the intended users. The application was then divided into functional modules so that different parts of the system could be developed and integrated. During development, the interface and backend were repeatedly modified as features were added, tested and improved.

4.8.1 Requirement Understanding

The initial requirement was to create a practical platform that could help people understand wildlife movement and potential conflict situations. From this requirement, the team identified the need for a map, analytics, incident reporting, risk prediction and a mechanism for updating datasets. The decision to support ten national parks also influenced the structure of the application because the platform needed to support park-specific information rather than a single fixed location.

4.8.2 Frontend Development

The frontend was developed as the main interaction layer of the platform. Major frontend modifications were part of my contribution. The interface was organized around the main functions of the project so that users could move from an overview dashboard to detailed map,

analytics, reporting, prediction and dataset-management screens.

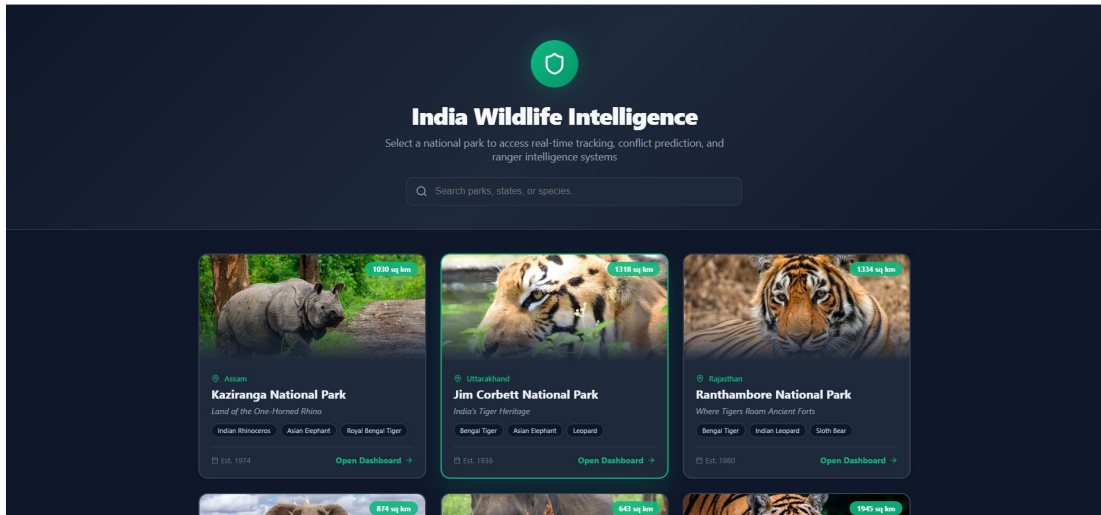


Figure 4.2: Dashboard / System Overview

The dashboard acts as an entry point to the system and gives the user access to the major functions. A consistent interface is important because the intended users may need to interpret several types of information during the same session.

4.8.3 Live Map and Wildlife Movement

The Live Map module is intended to provide spatial awareness of wildlife activity. It combines park information with movement or location indicators and conflict-related zones. The purpose of this view is to make spatial patterns easier to understand than a table of raw records.



Figure 4.3: Live Wildlife Map Interface

The map can be used by rangers to identify locations that may require attention and by community-oriented users to understand areas where wildlife activity is being represented by the system. The map is a decision-support interface; actual field conditions and official instructions remain important when making operational decisions.

4.8.4 Analytics

The Analytics module converts stored information into visual summaries. The demonstrated interface includes information related to wildlife sightings, threat severity and activity patterns. Such visualizations can help users identify trends and compare the relative importance of different categories of information.

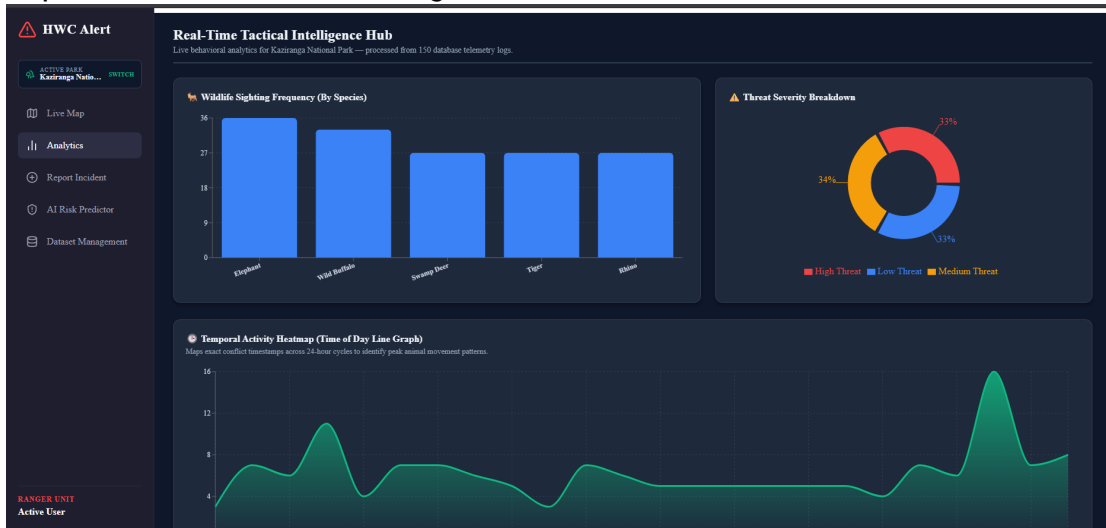


Figure 4.4: Analytics Module

Analytics are useful because raw datasets can be difficult to interpret quickly. By presenting summarized information, the platform can help a user identify patterns that deserve further investigation.

4.8.5 Incident Reporting

The Incident Reporting module provides a structured mechanism through which a wildlife sighting or conflict event can be entered into the system. The reporting interface is intended to standardize the information being submitted so that incident records can be reviewed alongside other wildlife information.

The screenshot shows the 'HWC Alert' interface with the 'Report Incident' option selected in the sidebar. The main content area is titled 'Dispatch Field Incident' and includes the instruction: 'Immediately broadcast a live human wildlife conflict sighting within Kaziranga National Park to the emergency map and analytics engine.' The form contains three sections: 'WILDLIFE SPECIES SIGHTED' with a dropdown menu showing 'Indian Rhinoceros'; 'THREAT / SEVERITY LEVEL' with a dropdown menu showing 'High (Immediate Danger / Village Intrusion)'; and 'KAZIRANGA NATIONAL PARK INCIDENT LOCATION' with a dropdown menu showing 'Kohora Range (Central)' and a 'Use Live GPS' button. A large red button at the bottom is labeled 'BROADCAST INCIDENT REPORT'. The bottom left corner indicates 'RANGER UNIT Active User'.

Figure 4.5: Incident Reporting Interface

A structured incident record can contribute to the overall information base of the platform. When new records are added, they can support monitoring and later analysis, subject to the quality and completeness of the submitted information.

4.8.6 AI Risk Prediction

The AI Risk Predictor is the decision-support component of the platform. It presents risk-oriented information that can help users understand when and where wildlife conflict may require increased attention. The demonstrated interface includes hourly and monthly risk information, conflict-type breakdowns, hotspot-related information, movement direction or activity windows, and suggested ranger actions.



Figure 4.6: AI Risk Prediction Interface

The purpose of this module is to transform available wildlife information into an easier-to-understand risk view. The output should be interpreted as predictive or advisory information rather than as a guaranteed statement of future animal behavior.

4.8.7 Dataset Management

Dataset Management was designed to address the changing nature of wildlife information. National parks may provide new datasets as additional observations or records become available. Instead of requiring developers to manually edit the application whenever such data arrives, the system provides an interface for uploading and managing new datasets.

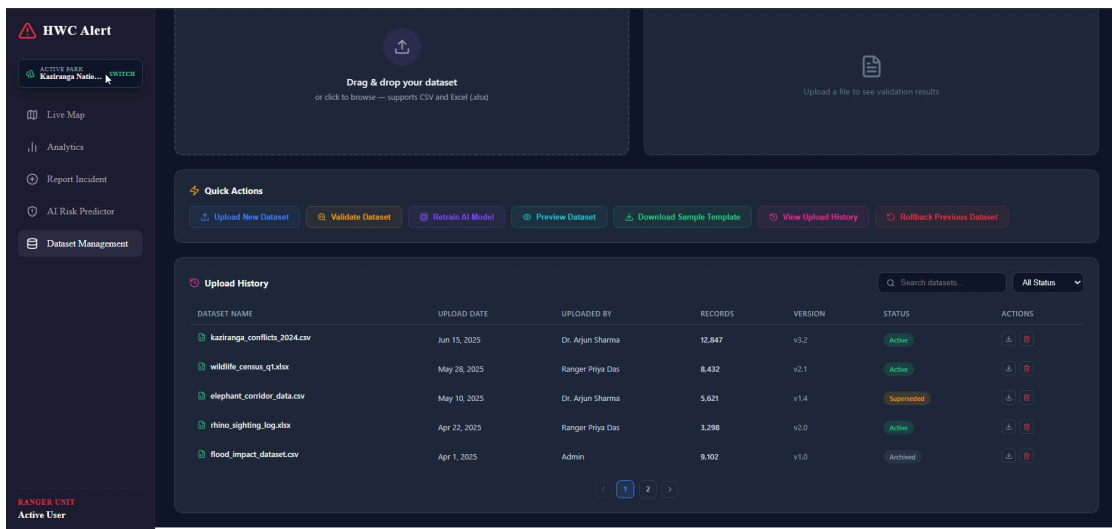


Figure 4.7: Dataset Management Interface

The demonstrated dataset workflow includes facilities for uploading data, checking the uploaded information, previewing it and maintaining versions or upload history. The interface also provides mechanisms related to managing changes to the dataset. This makes the data-management layer an important part of the project's long-term usability.

4.8.8 Multi-Park Operation

The platform was designed around ten national parks and provides options to switch between the supported parks. This allows the same application structure to be reused for different wildlife regions. Park selection is important because movement, incident and risk information must be associated with the correct geographic context.

4.8.9 Backend Contribution

My main technical contribution was concentrated on the backend and major changes to the frontend. The backend portion was important for connecting the user interface with the application's data and processing requirements. I also contributed to the overall system development and integration rather than treating the backend as an isolated component. Because the project was developed collaboratively, the final system should be understood as the combined result of the team. My report therefore records my substantial development involvement without attributing every module or every implementation decision solely to me.

4.8.10 Testing and Iterative Improvement

The system was reviewed through repeated demonstrations and iterative changes. During development, individual functions were checked from the user interface and the team worked on improving the connection between different modules. This process was particularly important because the project combined several different types of

functionality: mapping, analytics, reporting, prediction and data management.

The development process also highlighted an important practical aspect of software projects: a feature appearing on the interface does not necessarily mean that the underlying function is complete. Integration, data flow and interaction between modules must also be checked. This was an important part of the project-development experience.

4.9 Expected Outcomes

The expected outcome of the project is a centralized platform that makes wildlife movement and conflict-related information easier to access and interpret. The system is intended to support awareness and decision-making rather than replace professional wildlife management practices.

Outcome	Description
Centralized Information	Wildlife movement, incident, analytics and risk information are presented through one platform.
Improved Awareness	Users can obtain a clearer picture of wildlife activity and conflict-prone areas.
Ranger Support	Rangers can use maps, analytics and risk information as supporting information for planning and monitoring.
Community Support	Rural communities can gain access to information that may help them understand potential wildlife activity.
Data Updatability	New park datasets can be introduced through the Dataset Management module.
Multi- Park Scalability	The same platform structure can support information from multiple national parks.
Decision Support	AI-oriented risk information can help highlight periods, areas and categories requiring attention.

Future Extensibility	The modular design provides a basis for adding further parks, data sources, features and improved predictive methods.
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From an internship perspective, the project also provided experience in participating in a complete software project rather than working only on a small isolated exercise. Although the internship did not function as a formal technical training program, the project required practical coordination, implementation, modification, debugging and integration within a team.

4.10 Certificates of Completion and Other Communication Mail

Proof

The original internship completion certificate issued by MARS Internship should be attached in this section along with any relevant communication mail or proof requested by the university. The completion certificate is currently represented by the placeholder page earlier in this report because the scanned certificate was not provided while preparing this document.

Before final submission, the following should be inserted or attached as applicable:

- Scanned copy of the MARS Internship Completion Certificate.
- Any official internship communication mail required by the university.
- Any other proof specifically requested by the faculty coordinator.

The university's submission instructions require the Internship Report PDF, Internship Presentation PDF and Internship Completion Certificate PDF to be included in the public GitHub repository used for evaluation.

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

1. IILM University, "Internship Report Format updated 26-27," School of Computer Science and Engineering, IILM University, Greater Noida, 2026.
2. IILM University, "Submission of Summer Internship data for the AY 2026-27," communication circulated to students, 29 July 2026.
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